DPP	DEPARTMENT POLICY AND PROCEDURE		
Title	**RECOGNIZING ARRYTHMIAS IN PEDIATRICS**		
Department	**Pediatric Intensive Care Unit**		
Applies to	**All PICU Medical and Nursing Staff**		
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	NAME	POSITION	SIGNATURE	DATE
Prepared by:	**MS. REMYA MOHANDAS**	Quality Representative, Pediatric Intensive Care Unit		23/11/25
	MS. SARA ZAIBI AL SHOMRANI	Head Nurse, Pediatric Intensive Care Unit		23-11-25
	DR. LOTFI DJAIDER	Pediatric Consultant		23-11-25
Reviewed by:	**DRA. FAWZIA MOHD FRAHAT**	Head of Dept /Consultant , Pediatric Intensive Care Unit		7-12-25
	MR. MANSOUR AHMED AL-ZAHRANI	Head, Quality Management and Patient Safety		7-12-25
Approved by:	**MR. ABDULMAJEED MOHD AL-SOHEIMI**	Nursing Director		11-12-25
	DR. YOUCEF SOUAYEH	HED Assistant for Medical Affairs		11.12.25

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Baha_Cluster
Healthcluster-baha@moh.gov.sa
مستشفى المخواة العام
Al-Mikhwah General Hospital

1.0 PURPOSE

1.1. To unify the care of pediatric patients suffering from cardiac arrhythmias.

1.2. To enable all nurses to respond appropriately to a patient experiencing cardiac arrhythmias.

1.3. To understand causes of arrhythmias and to provide supportive treatment.

2.0 DEFINITION/ABBREVIATION

2.1. **Cardiac Dysrhythmias/ Arrhythmias**- are disturbances in regular heart rate and / or rhythm due to change in electrical conduction or automaticity. It may arise from the sinoatrial node or anywhere within the atria or ventricles. Some may be benign and asymptomatic, whereas other dysrhythmias are life-threatening.

2.1.1. Causes are as follows:

2.1.1.1. Benign Arrhythmias- include any transient episode of sinus bradycardia and tachycardia

2.1.1.2. Pathologic Causes

2.2. **Supraventricular Tachycardia (SVT)** - paroxysmal atrial tachycardia: a condition in which the heart rate speeds up due to a series of early beats from the sinus node or another pacemaker site above the ventricles. PAT usually begins and ends rapidly, occurring in repeated periods. This condition can cause symptoms such as weakness, fatigue, dizziness, fainting, or palpitations if the heart rate becomes too fast. The most common type of cardiac arrhythmia is seen in the neonate.

2.3. **Atrial Flutter** – It is difficult to distinguish from other supraventricular tachycardia. The number of P waves between each QRS complex is usually a constant number and is expressed as a ratio (i.e., a two-to-one atrial flutter means that two P waves are occurring between each QRS).

2.4. **Atrial Fibrillation** – is a condition in which the electrical signals come from the atria at a very fast and erratic rate. The ventricles contract in an erratic manner because of the erratic signals coming from the atria. It is less common than supraventricular tachycardia or atrial flutter.

2.5. **Wolf - Parkinson White Syndrome** – (short PR interval and delta wave, shows upstroke of the QRS complex) difficult to identify when the rate is fast. Ectopic beats

2.6. **Ventricular Tachycardia (VT)** – is a condition in which an electrical signal is sent from the ventricles at a very fast and even rate. If the heart rate is sustained at a high rate, symptoms such as weakness, fatigue, dizziness, fainting, or palpitations may be experienced.

2.7. **Atrioventricular block** - with symptoms occurring in newborn with complete AV block ventricular rates less than 55/min. secondary to extra cardiac disease.

2.7.1. Sepsis

2.7.2. Disease of the central nervous system

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2.7.3. Hypoglycemia

2.7.4. Drug toxicity

2.7.5. Adrenal Insufficiency

2.7.6. Electrolyte abnormalities

2.7.7. Metabolic acidosis or alkalosis

3.0 RESPONSIBILITY

3.1 **Physician** is responsible for interpreting heart rhythm and ordering appropriate emergency interventions. Responsible for performing cardioversion.

3.2 **Head Nurse** is responsible to evaluate staff nurses about their knowledge recognizing arrhythmia and emergency interventions in pediatric patients.

3.3 **Charge Nurses** are responsible to make sure that all orders are carried out and done in proper way and to help staff nurses in emergency cases.

3.4 **Staff nurses** are responsible to carry out orders, assess and re-assess patient and interpret heart rhythm. Responsible in notifying the physician if such arrhythmias occur. Must be knowledgeable with appropriate emergency interventions for pediatric patients with arrhythmias.

4.0 POLICY

4.1. PICU staff must have knowledge in ECG interpretation.

4.2. PICU staff must be able to respond appropriately to patient experiencing cardiac arrhythmias.

4.3. Crash cart and resuscitation equipment should be available in the bedside.

4.4. Staff nurse should assist the doctor in cardiopulmonary resuscitation

4.5. Intake and output must be properly monitored and IVF must be properly regulated.

5.0 PROCEDURE (S)

5.1. ***Preparatory phase***

5.1.1. Strict hand washing technique.

5.1.2. Perform complete physical examination:

5.1.2.1. Check for tachypnea, rales on pulmonary examination, enlarge liver, and cardiomegaly

5.1.3. Assist in laboratory studies.

5.1.3.1. Electrolyte, calcium and magnesium levels.

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5.1.3.2. Blood gas levels.

5.1.3.3. Drug levels

5.1.3.4. Chest X-ray

5.1.3.5. Electrocardiography

5.1.4. Assist in the insertion of IV cannula and administration of IV fluids as ordered.

5.1.5. Attach to cardiac monitor and pulse oximeter.

5.1.6. If arrhythmia is detected, full electrocardiographic evaluation should be performed.

5.1.7. Any acid-base disorder, hypoxia, or electrolyte abnormality needs to be corrected.

5.1.8. Monitor ABG as ordered. Essential for monitoring condition Involving hypoxemia, CO2 retention and pH level alteration.

5.1.9. Administer oxygen according to doctor's order.

5.1.10. Monitor serum electrolytes. To provide correction if result is beyond normal limit.

5.1.11. Check rate of IV fluid, regulate as ordered. To prevent over infusion or under infusion of patient.

5.1.12. Measure and record intake and output. To maintain hydration status. Both hypovolemia and hypervolemia can result in increased heart rate.

5.1.13. Regulate and maintain normal body temperature.

5.1.14. Administer pain reliever as ordered. Any source of pain whether acute or chronic, can lead to stimulation of the sympathetic nervous system and can increase the heart rate.

5.1.15. Administer sedation as ordered.

5.2. ***Performance phase***

5.2.1. **VAGAL MANEUVER** - used for SVT

5.2.1.1. Be sure to support the patient's airway, breathing and circulation.

5.2.1.2. If possible, obtain a 12-lead ECG before and after the maneuver.

5.2.1.3. Record and monitor the ECG continuously during the maneuver.

5.2.1.4. If patient is stable and the rhythm does not convert - you may repeat the attempt.

5.2.1.5. If the second attempt fails- select another method or provide pharmacologic therapy.

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5.2.1.6. If the patient is unstable- attempt vagal maneuvers only while making preparations for pharmacologic or electrical conversions.

5.2.2. **CARDIOVERSION**

5.2.2.1. Place paddles at the apex (left lower chest in 5th intercostals space in the anterior axillary line) and base of the heart (right of median below clavicle).

5.2.2.2. Place a saline soaked gauze pad beneath each paddle to ensure good electrical conduction. The dose is 1-4 joules/ kg which should be increased 50 -100% each time an electrical change is delivered.

5.2.2.3. When cardioversion is used for infant with ventricular fibrillation, the synch must be switched off.

5.2.2.4. NOTE: Synchronized Cardioversion is used for SVT, Atrial Flutter and VT with a pulse

5.2.3. **PHARMACOLOGIC THERAPY**

5.2.3.1. Drugs used for SVT:

5.2.3.1.1. Adenosine

5.2.3.1.2. Amiodarone

5.2.3.1.3. Procainamide

5.2.3.1.4. Verapamil for children above 1 year

5.2.3.2. Drugs used for VT with a pulse:

5.2.3.2.1. Amiodarone

5.2.3.2.2. Procainamide

5.2.3.2.3. Lidocaine

5.2.3.3. Drug used for Torsades de pointes:

5.2.3.3.1. Magnesium Sulfate

5.2.3.4. Drugs used for SVT with abnormal/aberrant intraventricular conduction

5.2.3.4.1. Amiodarone

5.2.3.4.2. Adenosine

5.2.3.4.3. Procainamide

5.3. Explain to parents all the procedures done to the infant.

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5.4. Documents patient's vital signs, nursing measures rendered, medications administered procedures done, and patient's response to intervention.

6.0 EQUIPMENTS

6.1. Crash cart / resuscitation equipment

6.2. ECG machine

7.0 ATTACHMENT/FORMS

7.1. N/A

8.0 REFERENCES

8.1. MOH Guidelines for Pediatric Care

8.2. DPP-PICU-0030(V2) – Recognizing Arrhythmias in Pediatric – PMSH, 01/10/1441

8.3. Saudi Central Board for Accreditation of Healthcare Institution (CBAHI) PICU standards, 3rd version, January 2016

9.0 REVISION CLAUSE

9.1. Used the new logo for AL-Baha Health Cluster.

9.2. Modify the policy and procedure lay out, changed the content of the Header and footer

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